



5CS020

HumanComputer Interaction
Design and Usability Testing
Report

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Abstract

This design and usability testing report focuses on the evaluation and improvement of E-Patrika, a digital news application developed to provide users with localized and categorized news content, weather updates, and AI-assisted interaction. The app was tested with an emphasis on usability and accessibility to ensure a smooth and inclusive user experience. This report outlines the features offered by e-Patrika, covering everything from news categorization to the chatbot and language switch functionalities.

In addition, the report documents the entire design process, which includes brainstorming ideas, creating user personas, designing wireframes, user flows, wireflows, and task flows that guided the development of the app. Feedback gathered through user testing is analyzed in detail, highlighting user experience strengths and areas for improvement. Finally, a revised version of the design, incorporating the user feedback, is presented to demonstrate how iterative testing and design refinement enhanced the overall experience of e-Patrika.

Acknowledgement

I am truly grateful to our module leader, Dipesh Shrestha Sir, and our module tutor, Anish Maharjan Sir, for presenting us with such an engaging project and for their continuous guidance throughout every stage of its development. Their support played a key role in helping me complete this project on time. I would also like to express my appreciation to Herald College, Kathmandu, for providing the necessary resources and environment to carry out this project successfully. Finally, a heartfelt thanks to my fellow classmates for their constructive feedback, which greatly contributed to improving the overall quality of my work.

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Introduction:

E-Patrika is a digital news app designed to deliver local, national, and classified news in a structured and user-friendly manner. The app aims to provide a smooth and informative experience through real-time updates, weather information, and AI chatbot support. It caters to users who want quick access to headlines, breaking news, and useful tools all in one platform.

E-Patrika organizes content into categories like politics, entertainment, sports, and technology, making it easier for users to find content based on their interests. The app also includes multilingual support, allowing users to switch between Nepali and English, which helps make the app inclusive for a broader audience.

The top navigation bar includes links to user profiles, chatbot support, and weather updates, along with a search function for finding specific topics or articles quickly. Users can interact with the AI chatbot for news summaries or assistance with using the app. Accessibility features such as Dark Mode, readable fonts, and a clear layout ensure that the app is usable for people with varying abilities and tech skills.

With its clean design, transparency in content sources, and personalized recommendations, E-Patrika enhances the news reading experience, blending convenience, clarity, and cultural relevance.

2. Design Process:

Before diving into the actual design, several key steps were carried out as part of the design process. These included brainstorming, creating user personas, designing wireframes, and mapping out the wireflow, user flow, and task flow.

2.1 Brainstroming

Brainstorming played a crucial role at the beginning of the project. It involved generating various ideas for potential features that the mobile app should include. During this phase, I compiled a list of essential features that would enhance the app's functionality and then focused on developing the corresponding pages based on those features.

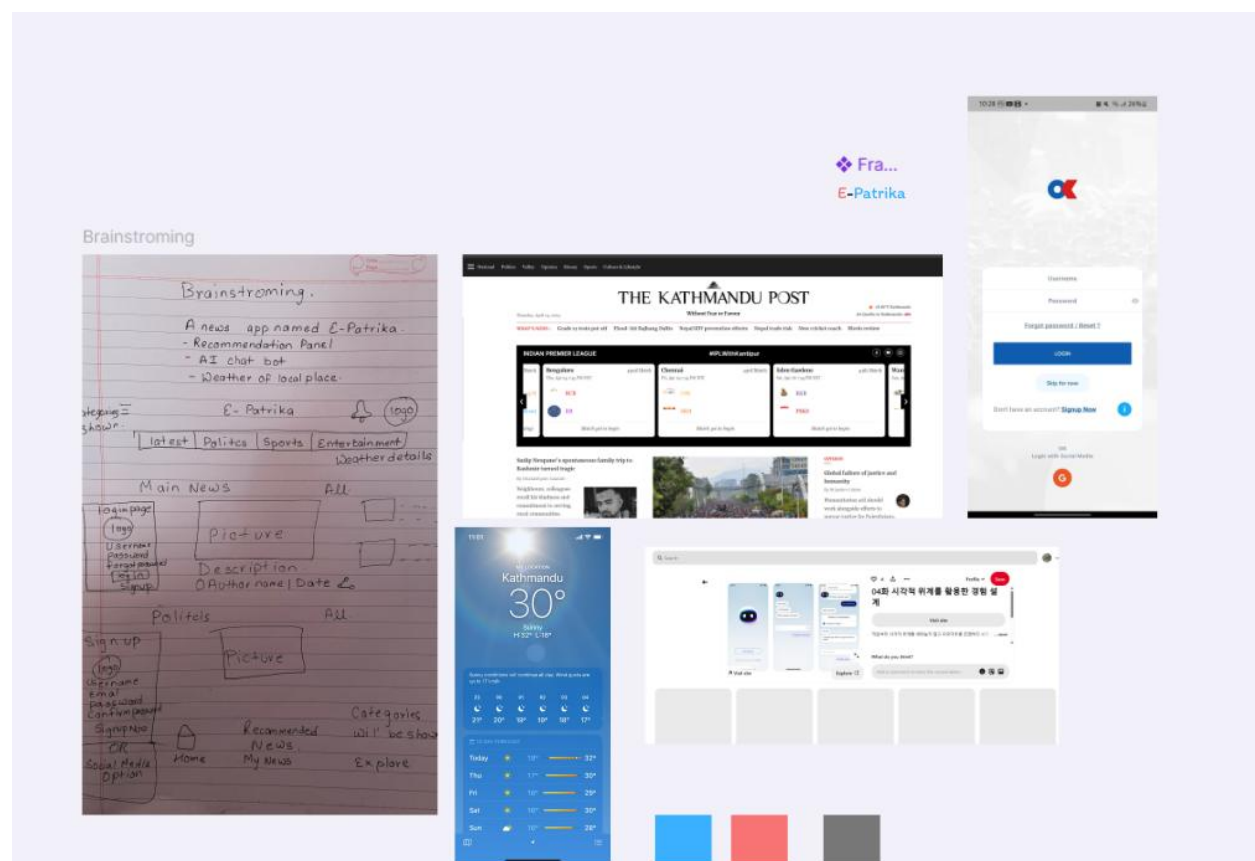


Figure 1 Brainstroming for E-Patrika

2.2 User Personas

User personas is the section which determines what kind of user uses my app. They find the problem of users and the app is build according to that . For this project I have made 2 user personas.

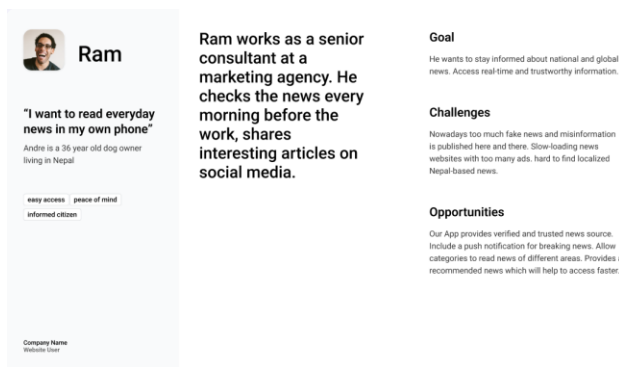


Figure 2User persona 1

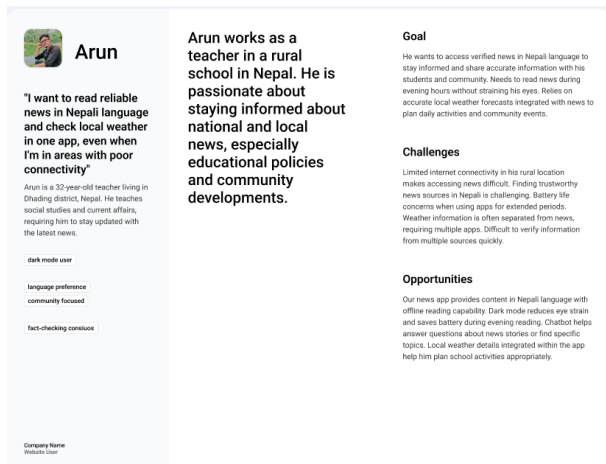


Figure 3User Persona 2

2.3 Userflow

Userflow helps us to find the how all the task is performed in the system including login signup to logout.

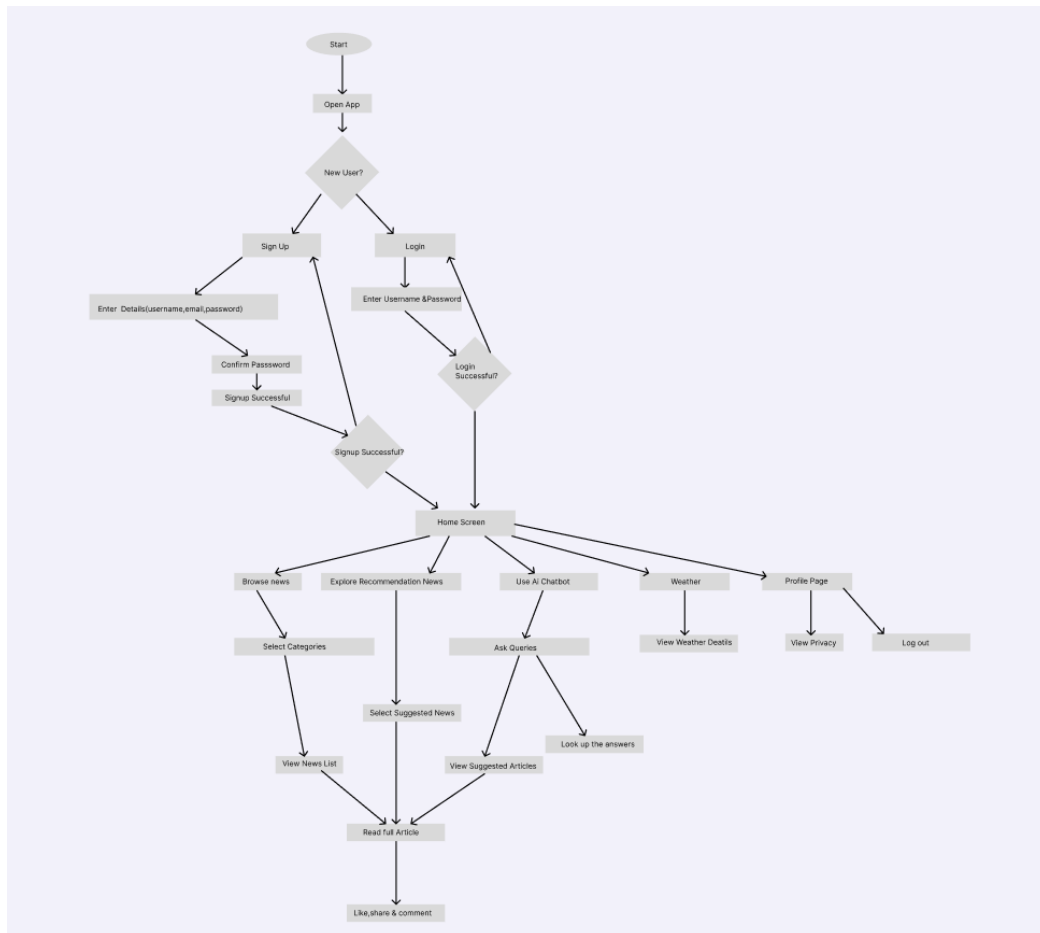


Figure 4 Userflow of E patrika

2.4 TaskFlow

Taskflow gives the flow of one features how is respective task conducted throught the system.

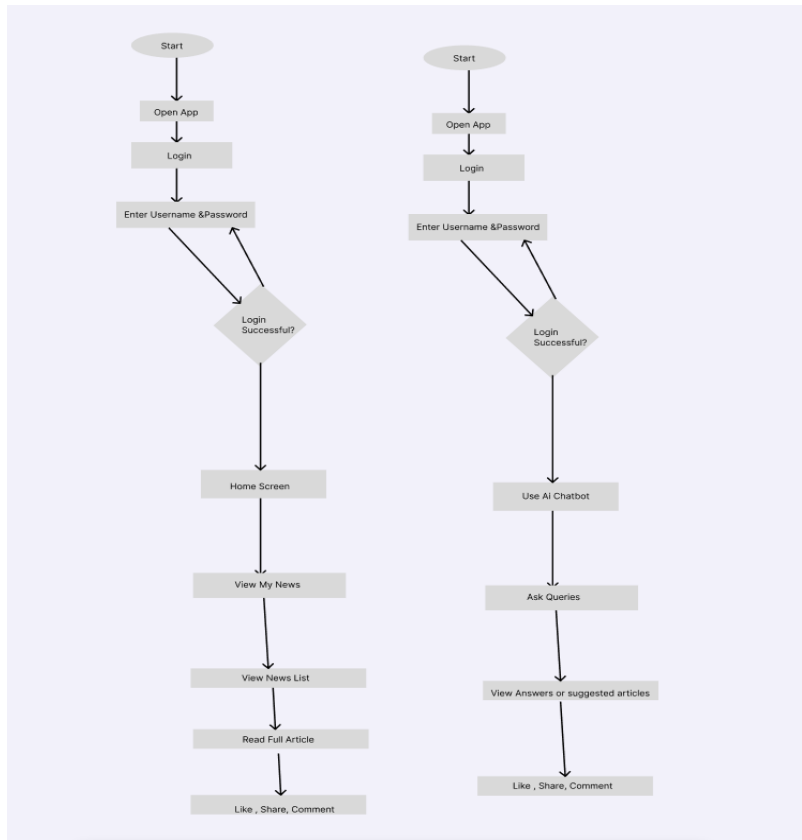


Figure 5 Task Flow

2.5 Wireframes

Designing the wireframe prior to starting the actual interface was incredibly helpful. It served as a clear roadmap throughout the entire app design process and significantly saved time. I created wireframes for all the key features planned for the mobile app, which helped me stay organized and focused. Additionally, working on the wireframes allowed me to improve and refine my overall design approach.

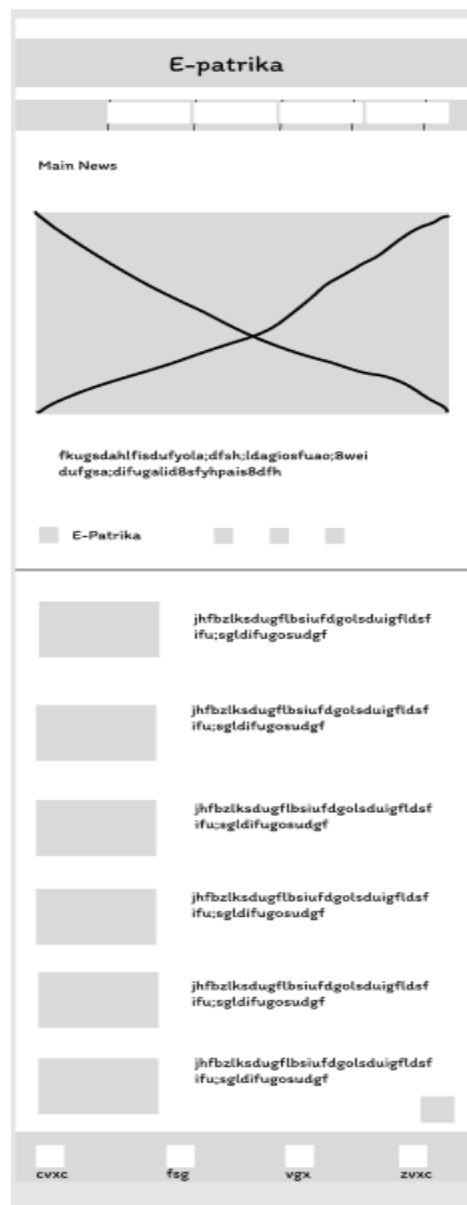


Figure 6 Wireframes of Landing page

2.6 Wireflow

Wireflow is the flow of process which helped me to design in a step by step process. Wire flow simply joins the line which indicates that when I click the specific place in the page what will come after that .

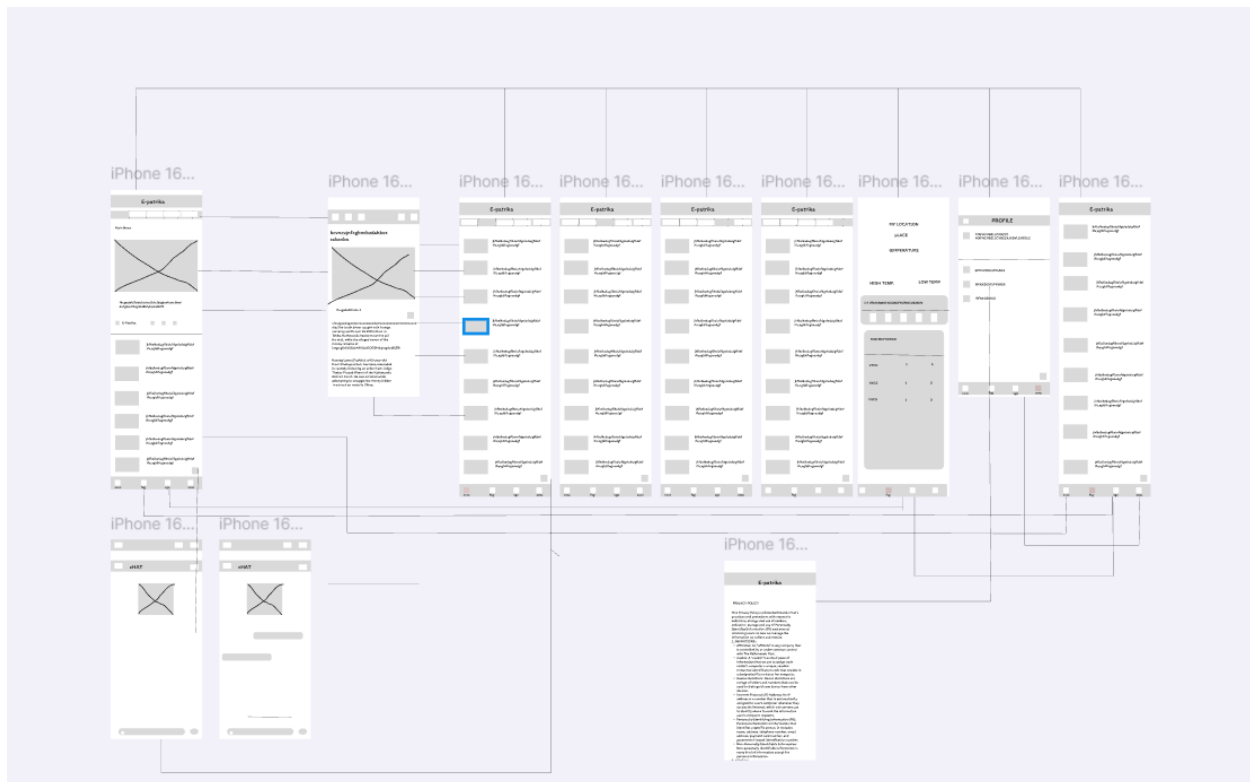


Figure 7Wireflow

2.7 Design Process

After all this features selection and pre design process, I started making design.

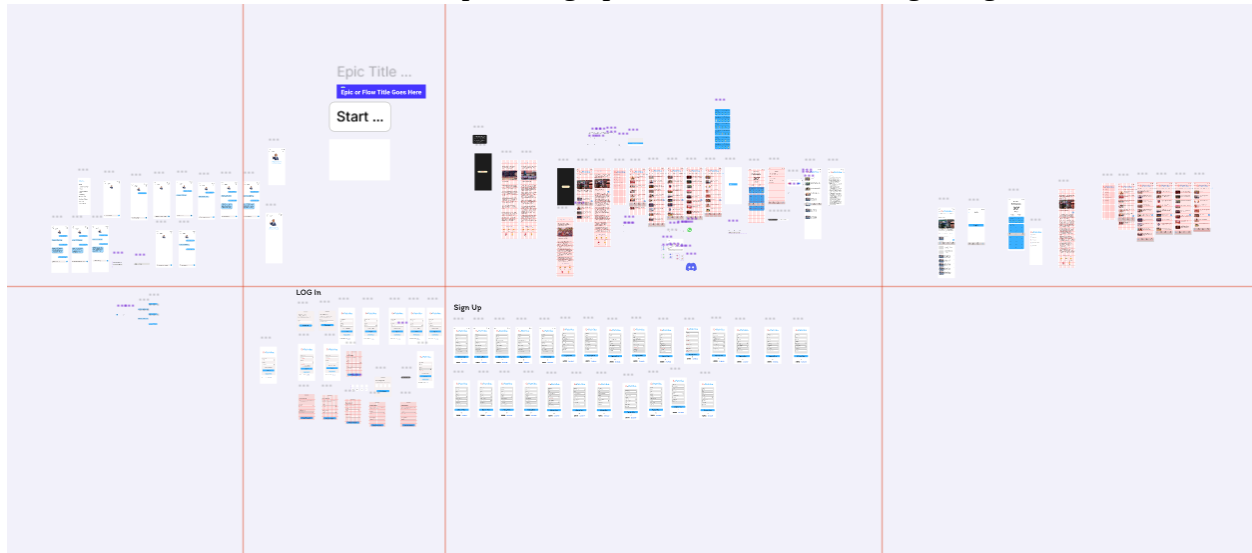


Figure 8 Design Process of E patrika

I designed all the screens for the mobile app **E-Patrika** based on the initial wireframes that were created earlier. I also added interactive elements like buttons and pop-ups to guide users and help them understand each step as they navigate through the app. I aimed to make the app feel like a real-life digital news platform.

For guidance and inspiration, I referred to various YouTube tutorials and websites that explained Figma tools and mobile design best practices. I kept the overall design simple and user-friendly. I carefully selected the color palette, fonts, and icons, and ensured consistency across all the screens to maintain a clean and professional look.

3. Evaluation and Testing

All mobile apps need to be tested to ensure they are working as expected and provide a smooth user experience. For testing, I first tested the **E-Patrika** app myself to identify any design or functionality issues. I made changes to the elements that I felt were missing or could be improved.

After that, I asked three of my friends to test the app for usability and accessibility. I observed how they interacted with the app to check whether the golden rules of interface design and accessibility standards were being followed. Their feedback helped me make the app more user-friendly and inclusive.

In my mobile app, I tried to incorporate most of the Shneiderman's eight golden rules and usability rules. I also included language translation feature and color change feature for accessibility.

3.1 Usability Evaluation

1. Strive for Consistency/Consistency and Standards: The icons, fonts and colors present in the website were consistent throughout all the pages. The placement of the navigation bar, header and the buttons were all consistent in my app.

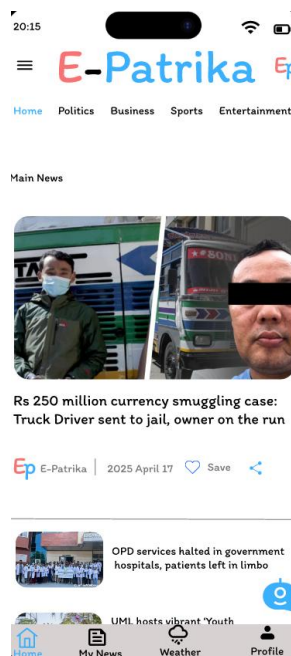


Figure 9Rule 1

2. Enable Frequent Users To Use Shortcuts/ Flexibility and efficiency of use

This rule was unfortunately not used in my project. There may be a search bar for searching news. I will redesign my app using this rule also.

Rule 3: Visibility of system status/ Offer informative Feedback: The app included pop ups to make the users aware of what steps they are currently on. While adding copying the news to share to others a popup comes that defines the link is copied and when logout is done by user a successful dialogue comes.

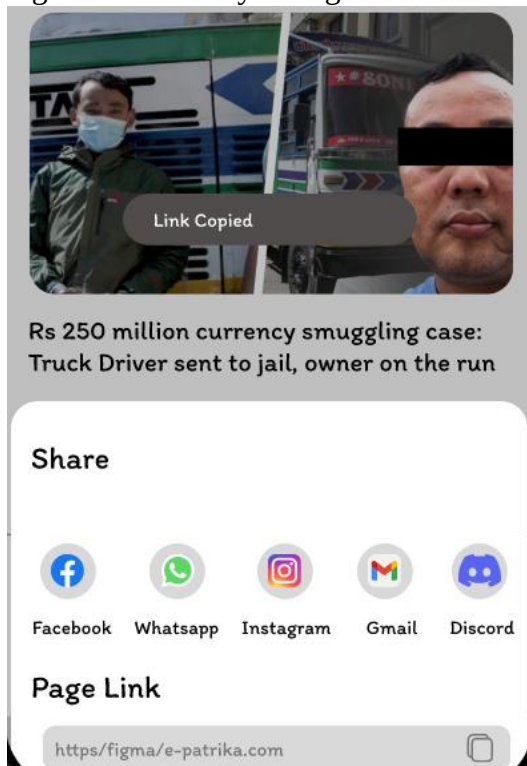
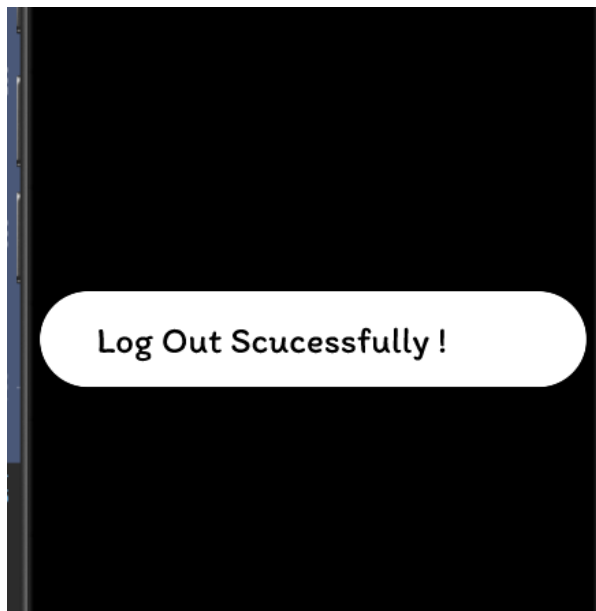


Figure 10 Rule 3



Rule 4: Offer simple error handling/ Help users recognize, diagnose, and recover from errors: Simple error handling is very important part of project so I have used it in login page. The password field in log in page provided error related to enter a field that is not filled to login.

E-Patrika

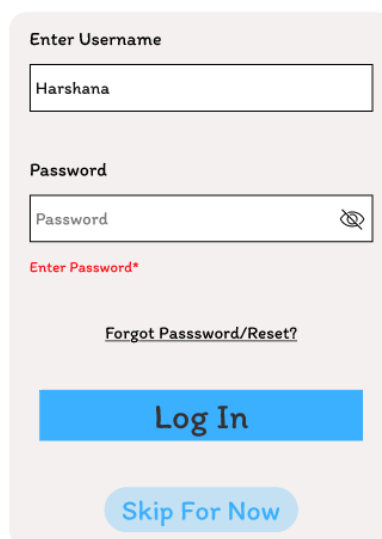


Figure 11Rule 3

Rule 5: Permit easy reversal of actions/User control and freedom: There was back button in the news details page when a news reading is completed they redirect back to the page and also in ai Chabot page there is back button to go to the home page.



Rs 250 million currency smuggling case: Truck driver sent to jail, owner on the run



Kathmandu April 17



The truck driver caught with foreign currency worth over Rs 250 million in Tokha, Kathmandu has been sent to jail for trial, while the alleged owner of the money remains at large.

Kusang Lama (Fuchhe), a 43-year-old from Sindhupalchok, has been remanded in custody following an order from Judge Thakur Prasad Kharel of the Kathmandu District Court. He was arrested while

Figure 12Rule 5



Rule 6: Reduce short-term memory load/ Recognition rather than recall: This feature was also not used in my project.

Rule 7: Dialogue should yield closure: To keep the user informed about the system status, a dialog pop-up was added during the logout process, allowing the user to choose whether to log out or remain signed in.

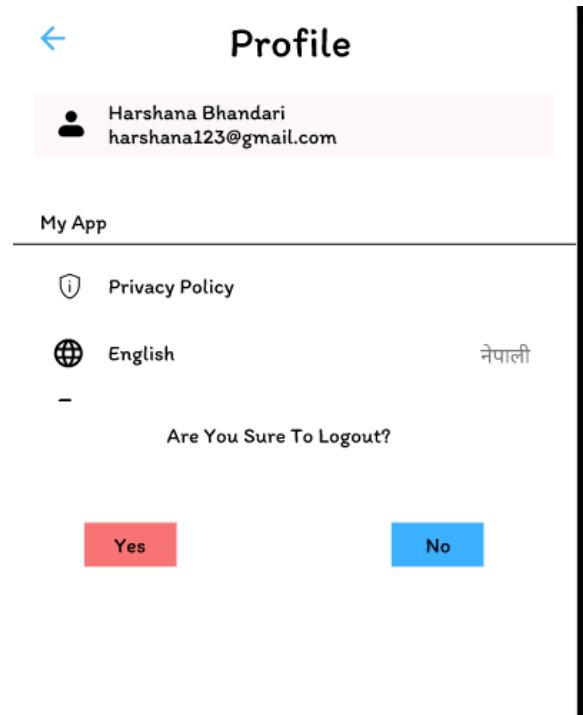


Figure 13 Rule 7

Rule 8: Support internal locus of control: The users can transfer from one page to next page easily as the can go to news by categories as home, politics, entertainment, weather page, my news and so on order to make them feel a sense of control over their actions.

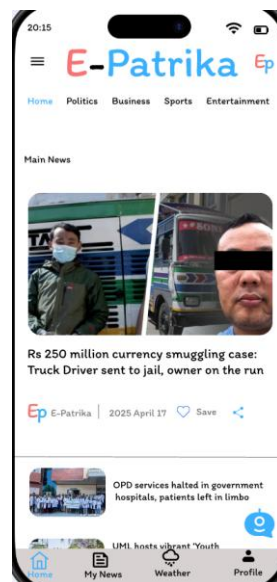


Figure 14 Rule 8

Rule 9: Match between system and real world: This app labels simple words like Home, My News, Politics, Entertainment, Business and so on which makes the app user friendly. There was no extra hard words that may be hard for the users to use the app beside news content.

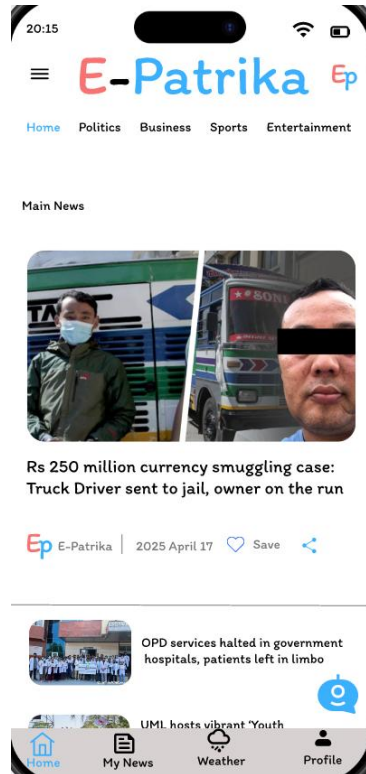


Figure 15 Rule 9

Rule 10: Aesthetic and minimalist design: The design in this app was simple and minimalist. being a news portal app the app does not contains some extra ordinary layout, design which makes user to read the content in easy way.

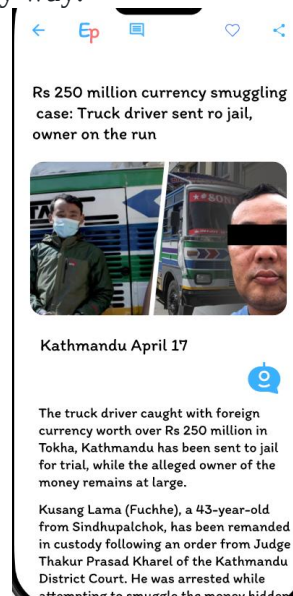


Figure 16 Rule 10

Rule 11: Help and documentation: In this app there is privacy and policy section but no proper help and documentation section so this need to be added.

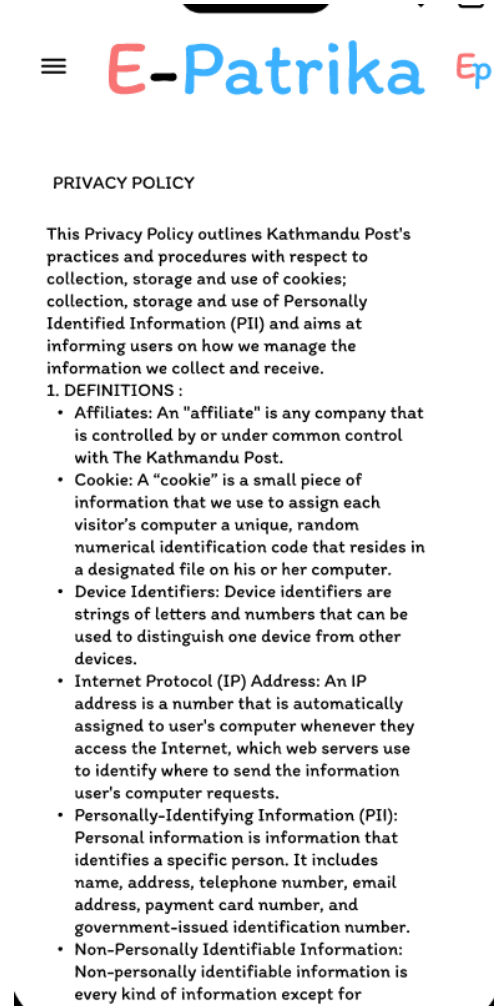


Figure 17 Rule 11

3.2 Accessibility Evaluation

When designing a mobile app, it's important to consider various factors and not just target a specific group of users. To ensure the app is accessible to a broader audience, I have included two key accessibility features.

3.2.1 Dark Mode Feature

User can change the app to dark mode so that it will be easier them to access the app in their own way. For this generation dark mode has become the basic need so I choose the dark mode as accessibility.



Figure 18 Accessibility Feature 1

3.2.2 Language Translate to Nepali

As my users are mostly from Nepal, it will be easy for them to read news in their own mother language. So I used the language in Nepali which ensures that the user who cannot read in English also access my app.

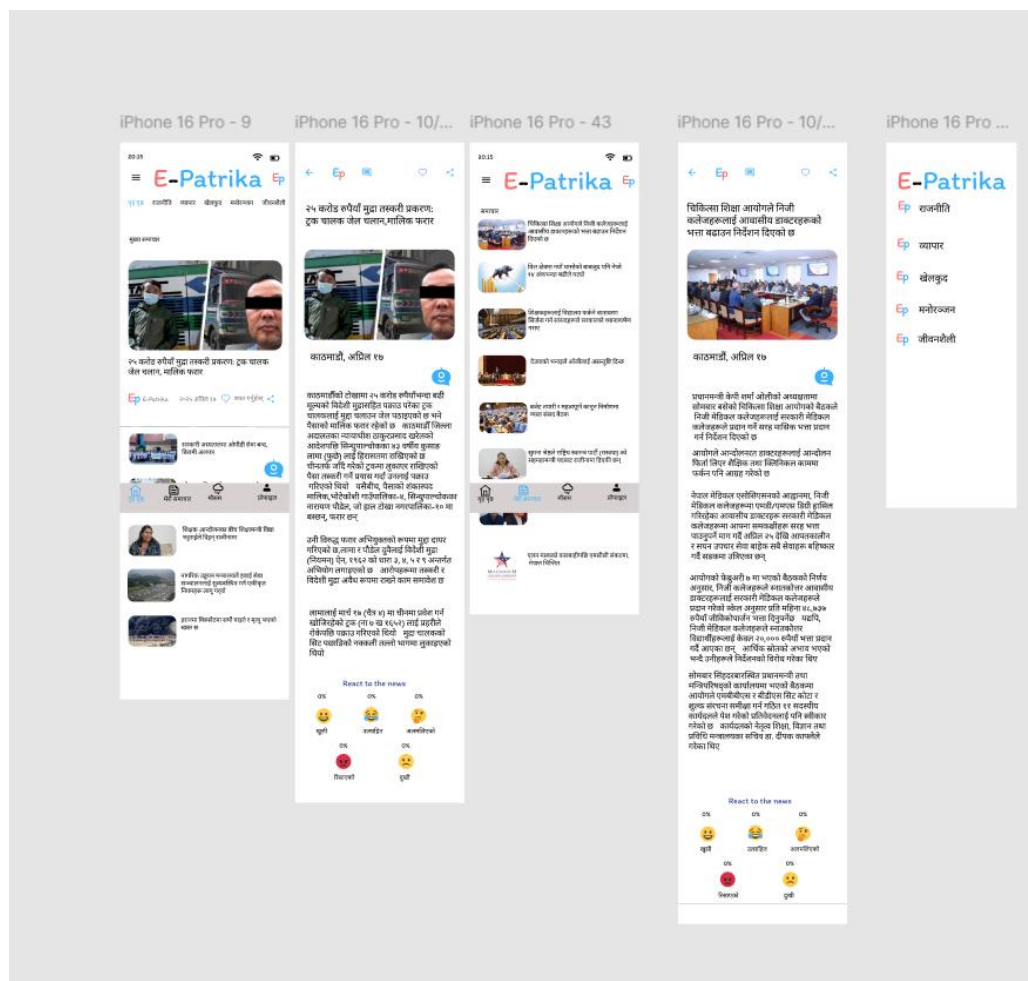


Figure 19 Accessibility Feature 2

3.3 Accessibility and Usability Questions for user testing

The user testing questions were shared through a Google Form, as it simplifies the process for users to submit their answers and makes response collection more efficient.

3.3.1 Questions

1. What is your name?
2. What is your age group?
3. What is your gender?
4. What type of news are you most interested in?
Usability
5. Do you find the color, fonts, and icons consistent throughout the app?
6. How would you rate the overall user interface (UI) of e-Patrika?
7. How easy was it to navigate between different pages (news, weather, chatbot)?
8. Which feature did you use the least and why?
9. Were there any confusing steps during navigation or interaction?

10. Do you feel a sense of control over your interactions while using the app?
11. How easy was it to find news articles of interest?
12. Was the layout of the articles (font size, image placement, line spacing) comfortable for reading?
13. Did the headlines and categories help you discover what you were looking for?
14. Did you use the local weather feature?
15. How accurate and useful did you find the weather data?
16. Would you like to see additional features such as forecasts or alerts?
17. Did you interact with the AI chatbot?
18. How helpful was the chatbot in answering your questions or guiding you through the app?
19. Was the chatbot's tone friendly and easy to understand?
20. Did the chatbot feel relevant and contextual to your needs?
21. Did the app give feedback after each action you took (like saving, loading, switching pages)?
22. Were you able to easily recover if you made a mistake (e.g., clicked the wrong page or option)?
23. Did the icons and labels in the app feel familiar and easy to understand?
24. Did you find any guidance (tooltips, instructions) helpful while using the app?

Accessibility

25. How much did the Dark Mode improve readability and reduce eye strain while reading articles?
26. How would you rate your overall experience with Dark Mode in terms of readability and comfort?
27. Did Dark Mode improve the visibility of key elements (e.g., text, buttons, images) on the website?
28. How clearly was the Nepali language version presented when you switched languages?
29. How would you rate the accuracy and readability of the Nepali translation in terms of accessibility for native speakers?
30. Do you think e-Patrika considers users with visual impairments (e.g., button size, color contrast)?
31. How likely are you to recommend e-Patrika to a friend or colleague?
32. Are the error messages in the app clear and helpful when something goes wrong?
33. What additional features would you like to see in e-Patrika?
34. Any other feedback or suggestions to improve your experience?

3.4 Testing Response

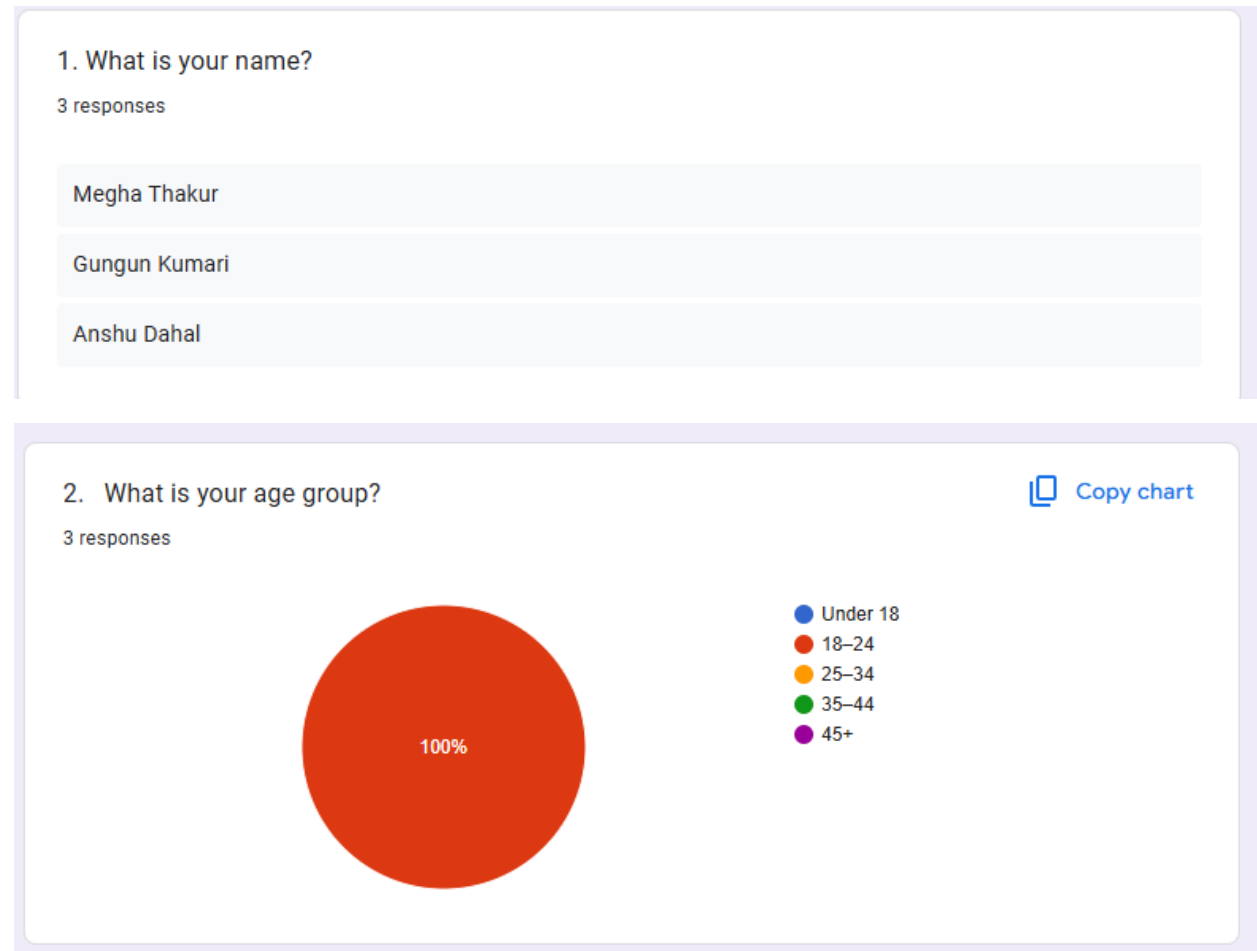
For testing I choose 3 of my friends:

Gungun Kumari

Megha Thakur

Anshu Dahal

Following Response were Collected:



3. What is your gender?

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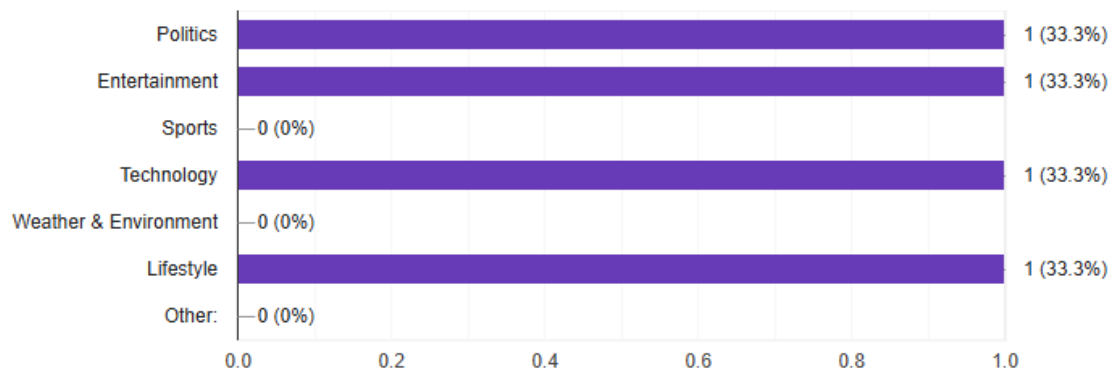
3 responses



4. What type of news are you most interested in?

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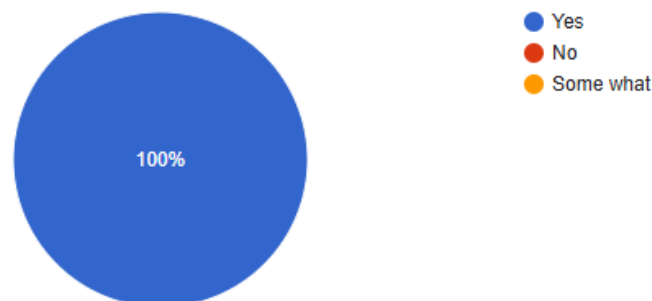
3 responses



5 Do you find the color, fonts, and icons consistent throughout the app?

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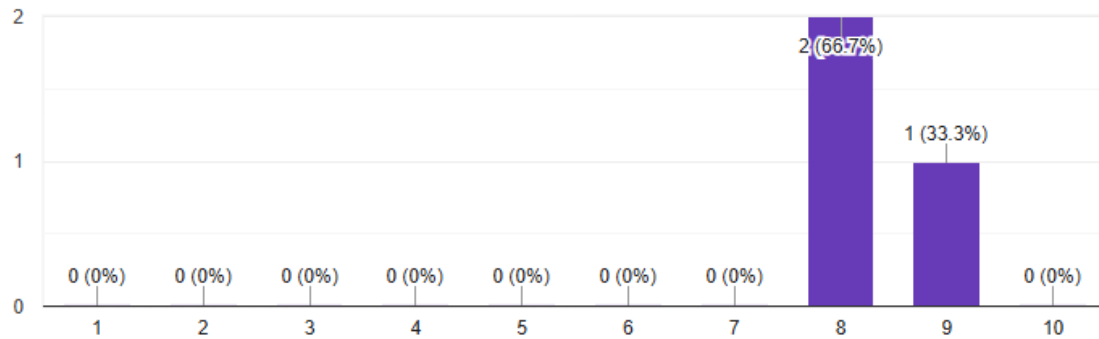
3 responses



6 How would you rate the overall user interface (UI) of e-Patrika?

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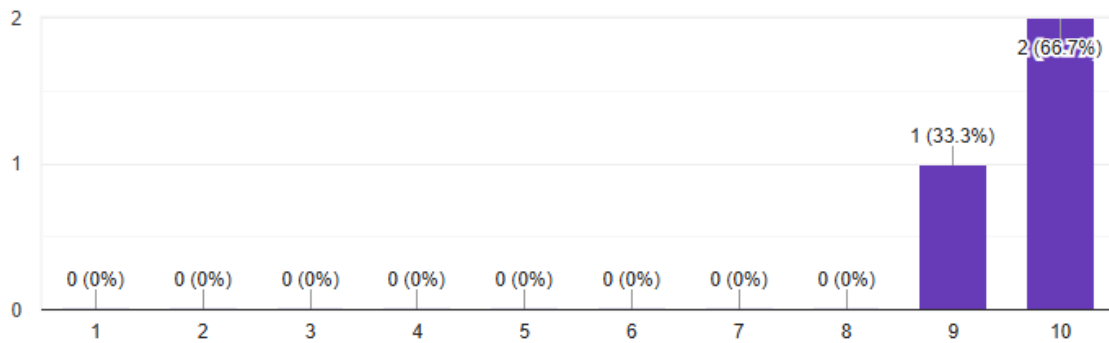
3 responses



7. How easy was it to navigate between different pages (news, weather, chatbot)?

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3 responses



8. Which feature did you use the most and why?

3 responses

Chatbot

Chat-bot

chatbot

9. Which feature did you use the least and why?

3 responses

Weather

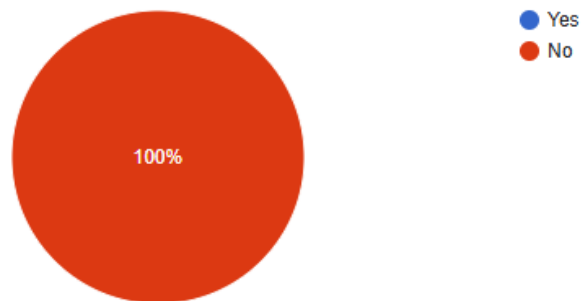
Sports because I am not interested in sports.

language translator

10. Were there any confusing steps during navigation or interaction?

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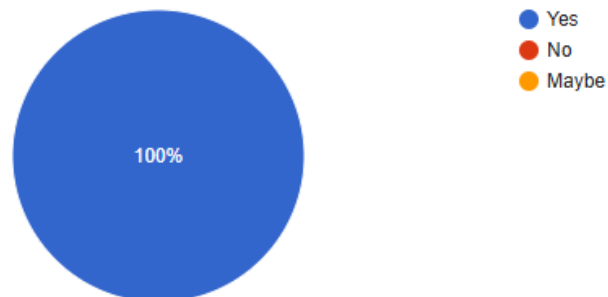
3 responses



11. Do you feel a sense of control over your interactions while using the app?

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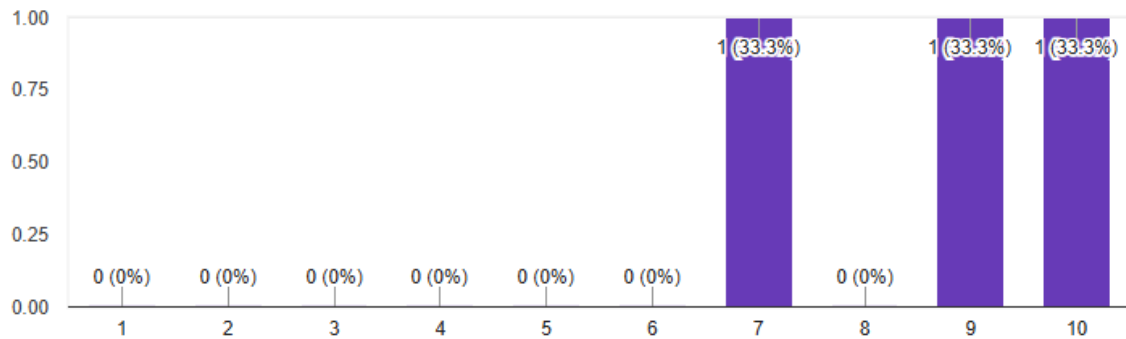
3 responses



12. How easy was it to find news articles of interest?

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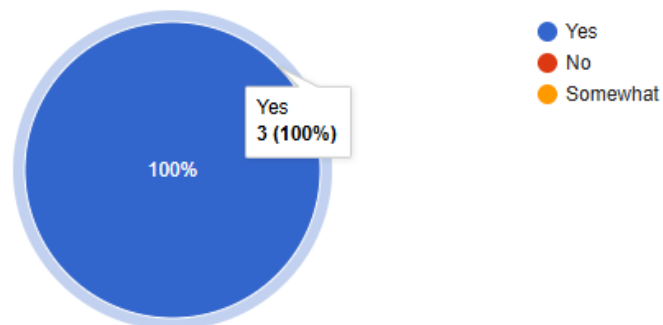
3 responses



13. Was the layout of the articles (font size, image placement, line spacing) comfortable for reading?

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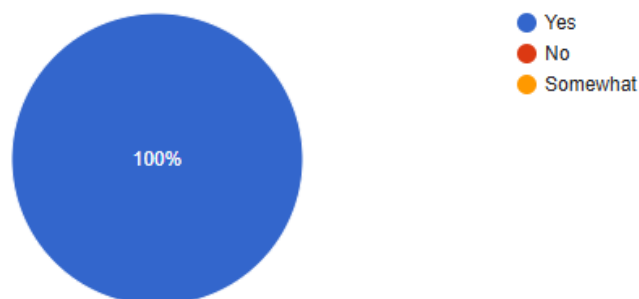
3 responses



14. Did the headlines and categories help you discover what you were looking for?

[Copy chart](#)

3 responses

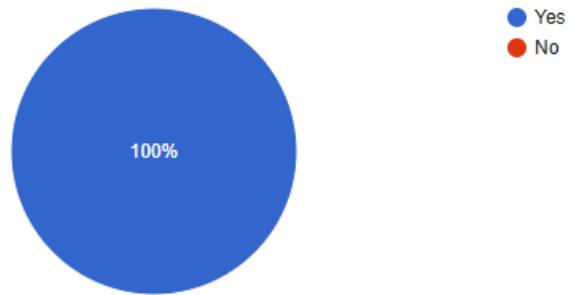


Weather Feature

15. Did you use the local weather feature?

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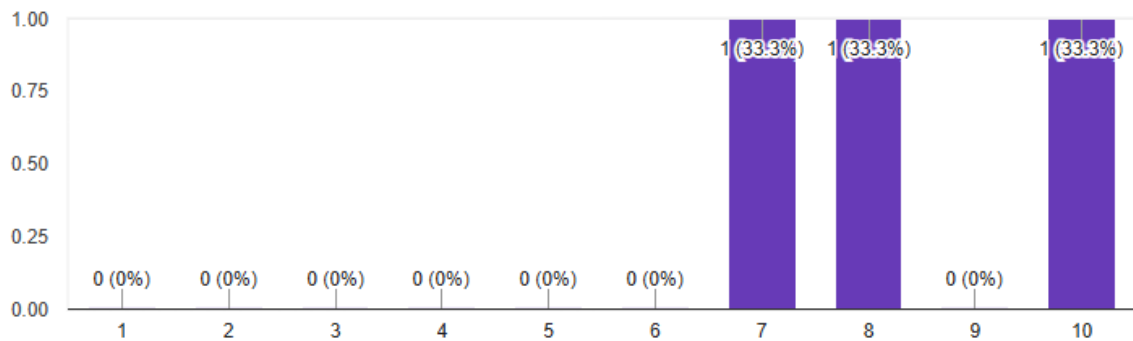
3 responses



16. How accurate and useful did you find the weather data?

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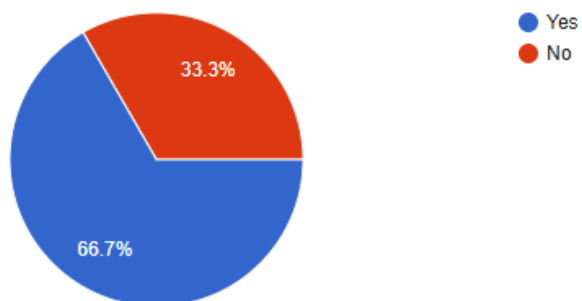
3 responses



17. Would you like to see additional features such as forecasts or alerts?

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3 responses

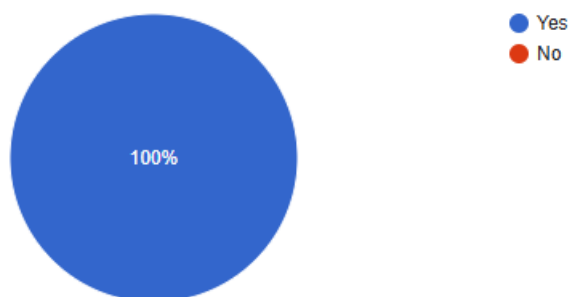


AI Chatbot Experience

18. Did you interact with the AI chatbot?

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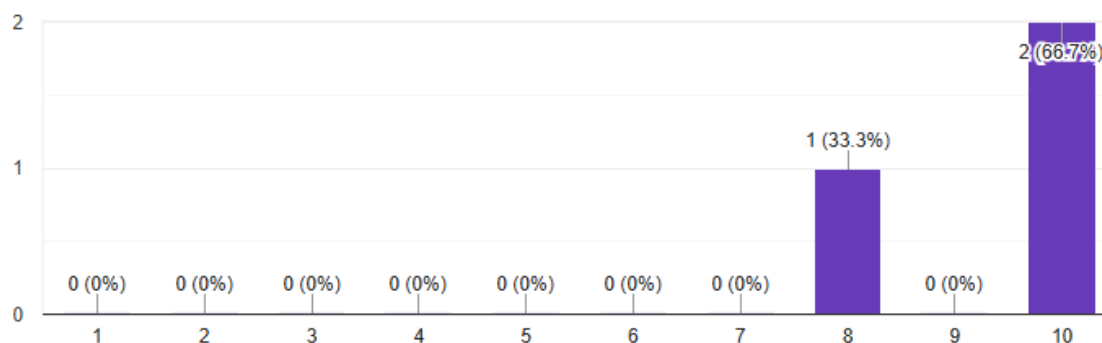
3 responses



19. How helpful was the chatbot in answering your questions or guiding you through the app?

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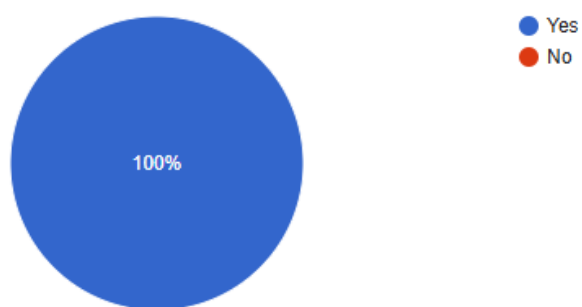
3 responses



20. Was the chatbot's tone friendly and easy to understand?

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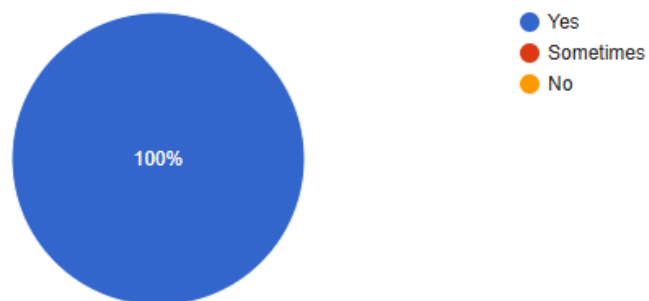
3 responses



21. Did the chatbot feel relevant and contextual to your needs?

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3 responses



22. Did the app give feedback after each action you took (like saving, loading, switching pages)?

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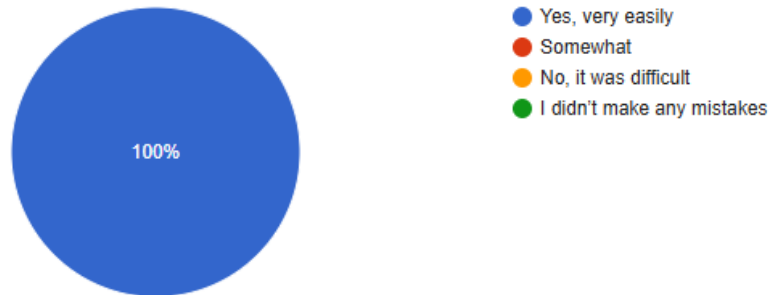
3 responses



23. Were you able to easily recover if you made a mistake (e.g., clicked the wrong page or option)?

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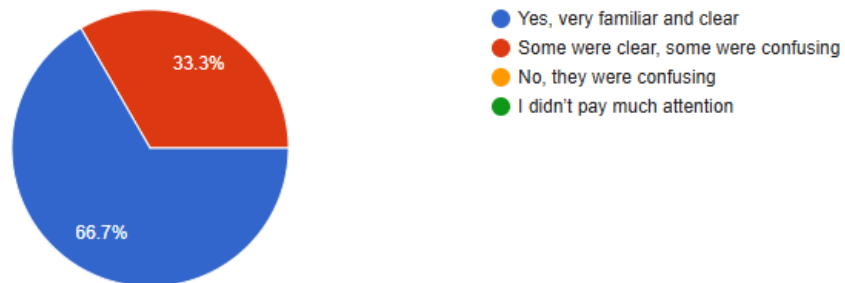
3 responses



24. Did the icons and labels in the app feel familiar and easy to understand?

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3 responses



25. Did you find any guidance (tooltips, instructions) helpful while using the app?

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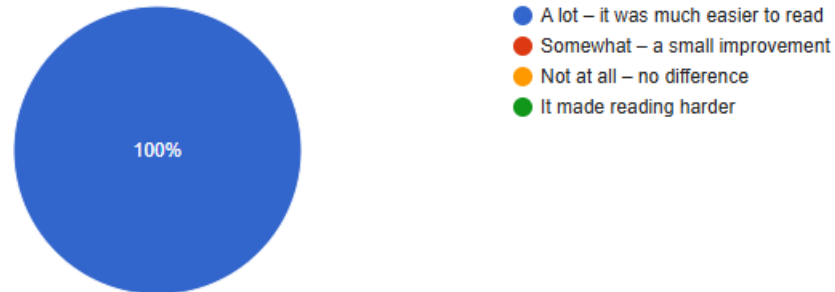
3 responses



26. How much did the Dark Mode improve readability and reduce eye strain while reading articles?

[Copy chart](#)

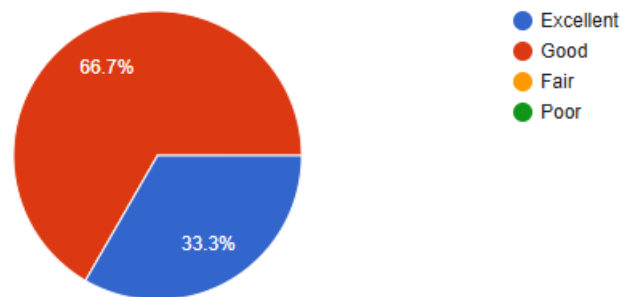
3 responses



27. How would you rate your overall experience with Dark Mode in terms of readability and comfort?

[Copy chart](#)

3 responses



28. Did Dark Mode improve the visibility of key elements (e.g., text, buttons, images) on the website?

 [Copy chart](#)

3 responses



- Yes, everything was clearer
- Some elements were easier to see
- No change
- Some elements became harder to see

29. How clearly was the Nepali language version presented when you switched languages?

 [Copy chart](#)

3 responses

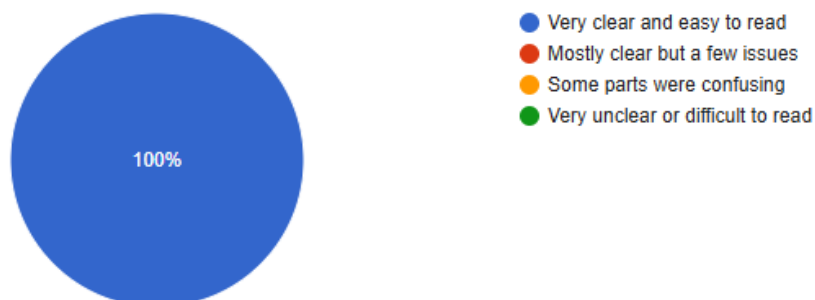


- Very clear and easy to read
- Mostly clear but a few issues
- Some parts were confusing
- Very unclear or difficult to read

29. How clearly was the Nepali language version presented when you switched languages?

 [Copy chart](#)

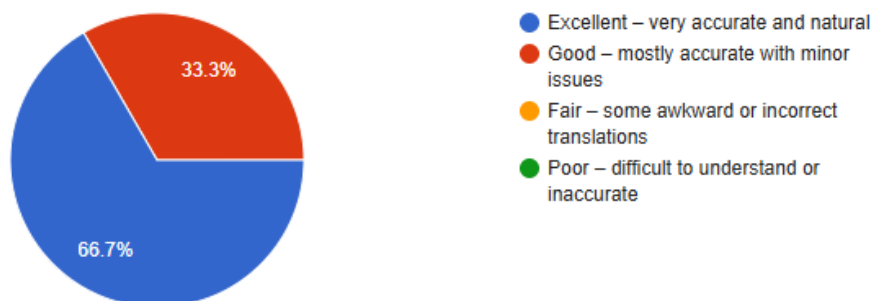
3 responses



30. How would you rate the accuracy and readability of the Nepali translation in terms of accessibility for native speakers?

 [Copy chart](#)

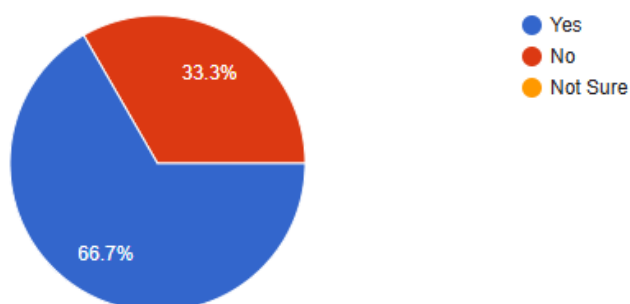
3 responses



31. Do you think e-Patrika considers users with visual impairments (e.g., button size, color contrast)?

 [Copy chart](#)

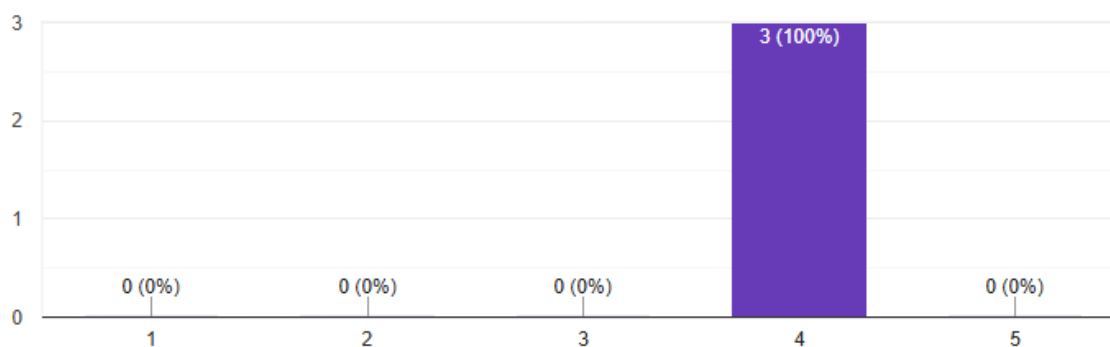
3 responses



32. How likely are you to recommend e-Patrika to a friend or colleague?

 [Copy chart](#)

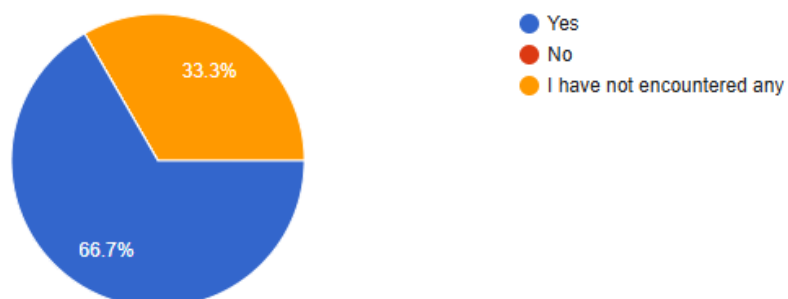
3 responses



33. Are the error messages in the app clear and helpful when something goes wrong?

 [Copy chart](#)

3 responses



34. What additional features would you like to see in e-Patrika?

3 responses

Add more chat in chatbot

Additional Feature must be that something like current update in the city should be shown.

forecast

35. Any other feedback or suggestions to improve your experience?

2 responses

no

Yes, she could access more weather details and make ui more attractive in her app.

Link to goggle drive containing testing Videos:

<https://drive.google.com/drive/folders/1bhKSFz8VZxSPcVOznqxMdbnBJpVkFmrA>

3.5 Revamped Design

After the suggestion of user who tested my app, I have made some changes to ensure good functionality.

First of all I added the more information in the weather section where more weather details can be seen.

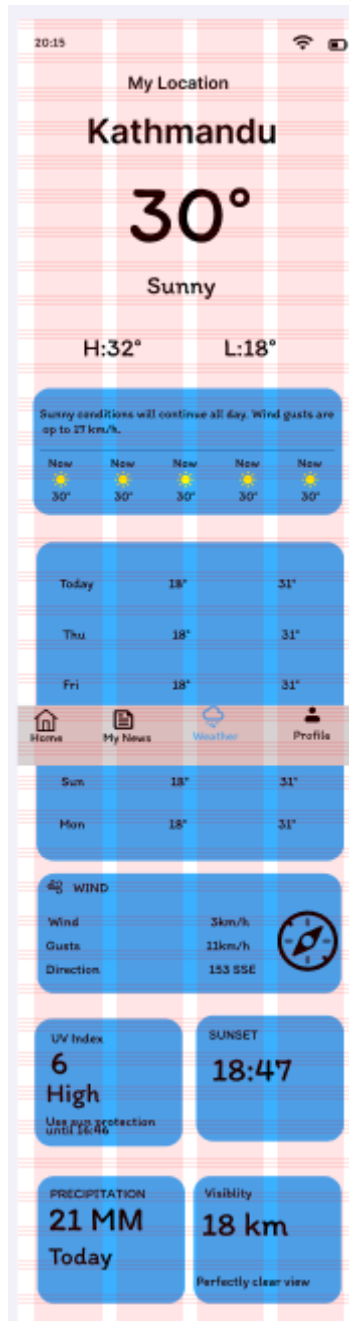


Figure 20 Revamped Weather page

Other changes I have done prototyping for all those buttons which was not working and some of the functionality of chat bot is also changed.

4. Conclusion

The overall design process of this mobile app has been a great learning experience for me. Through this task, I have significantly improved my skills in Figma. I've come to understand the various stages involved in designing a mobile application. In addition, I've gained important knowledge about the key principles that should be followed during the design process. I also learned the importance of accessibility and how many different factors need to be considered to make an app inclusive. Moreover, I now understand the value of user testing to gather feedback and improve the app based on real user input.

5. Appendix

5.1 Screenshot of all Design Process:

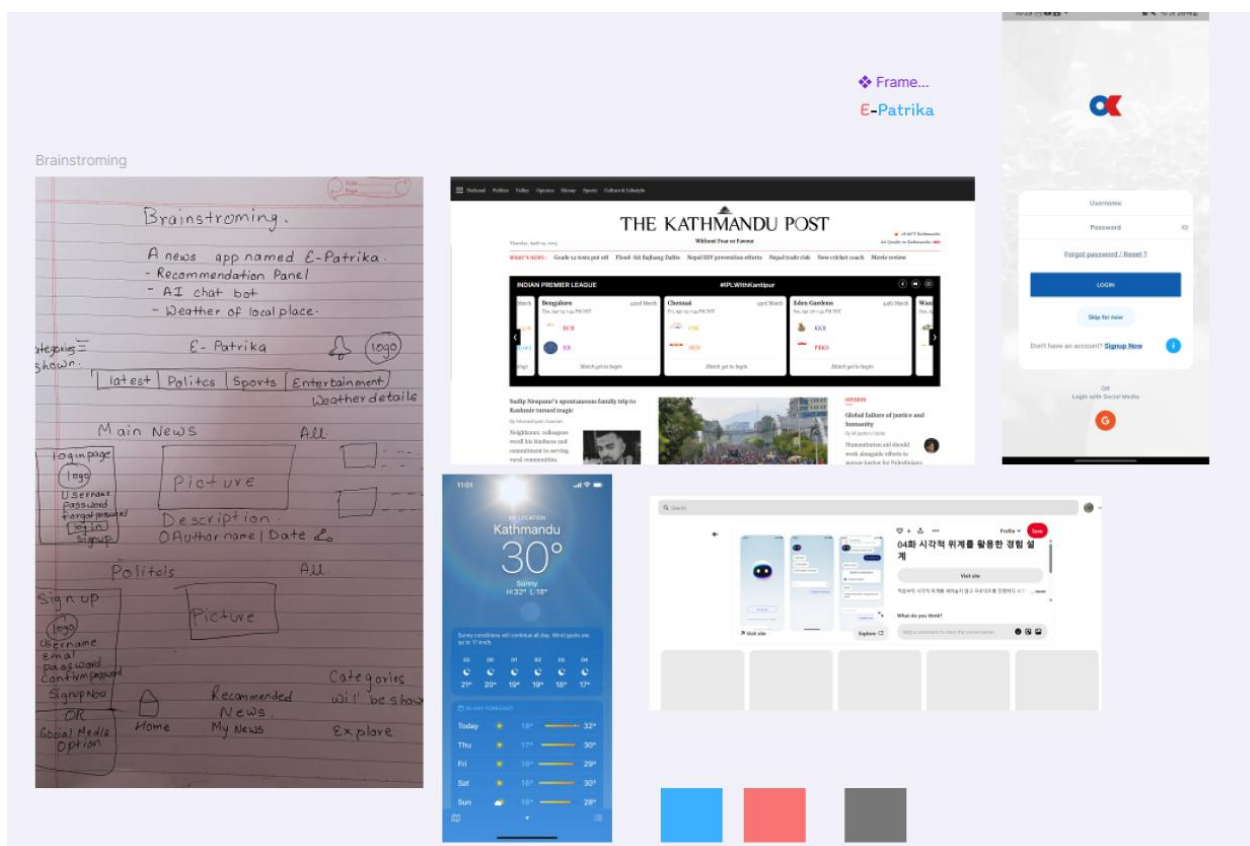


Figure 21 Brainstroming

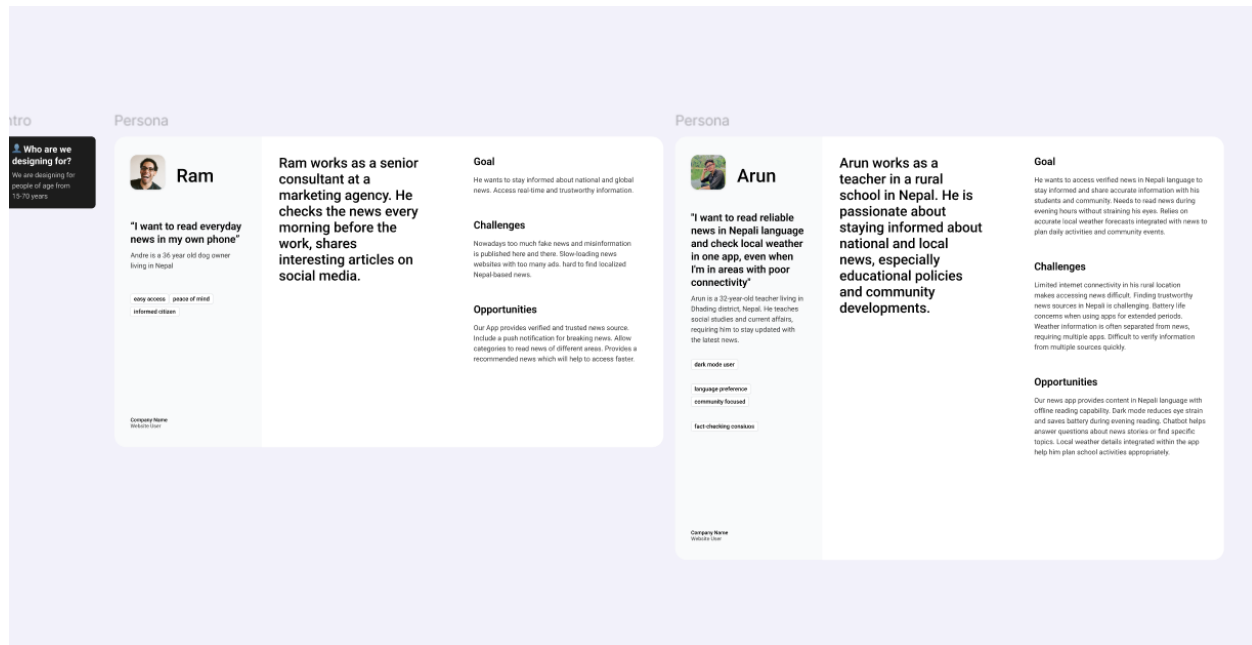


Figure 22 User personas of E-Patrika

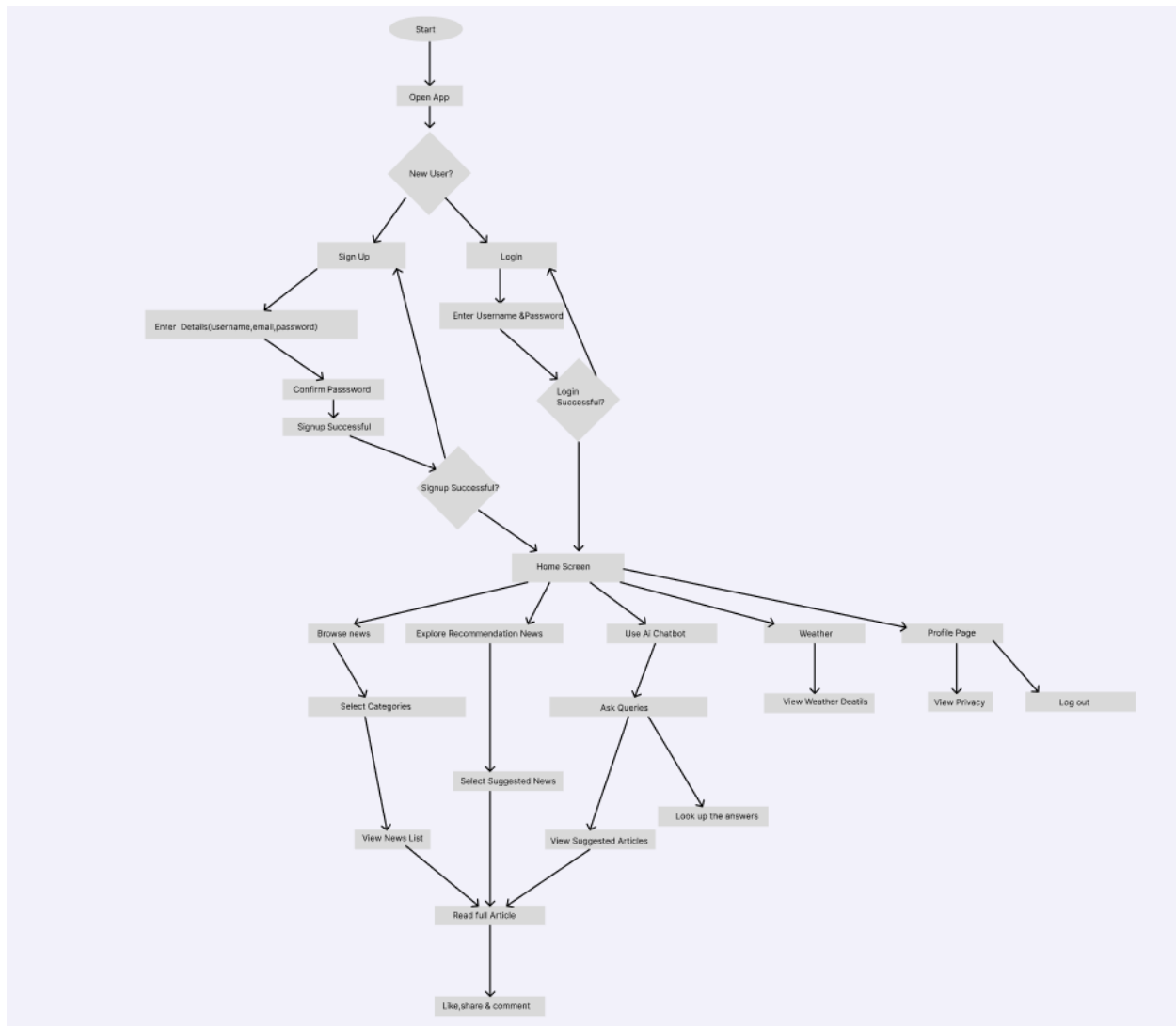


Figure 23 Userflow of E –Patrika

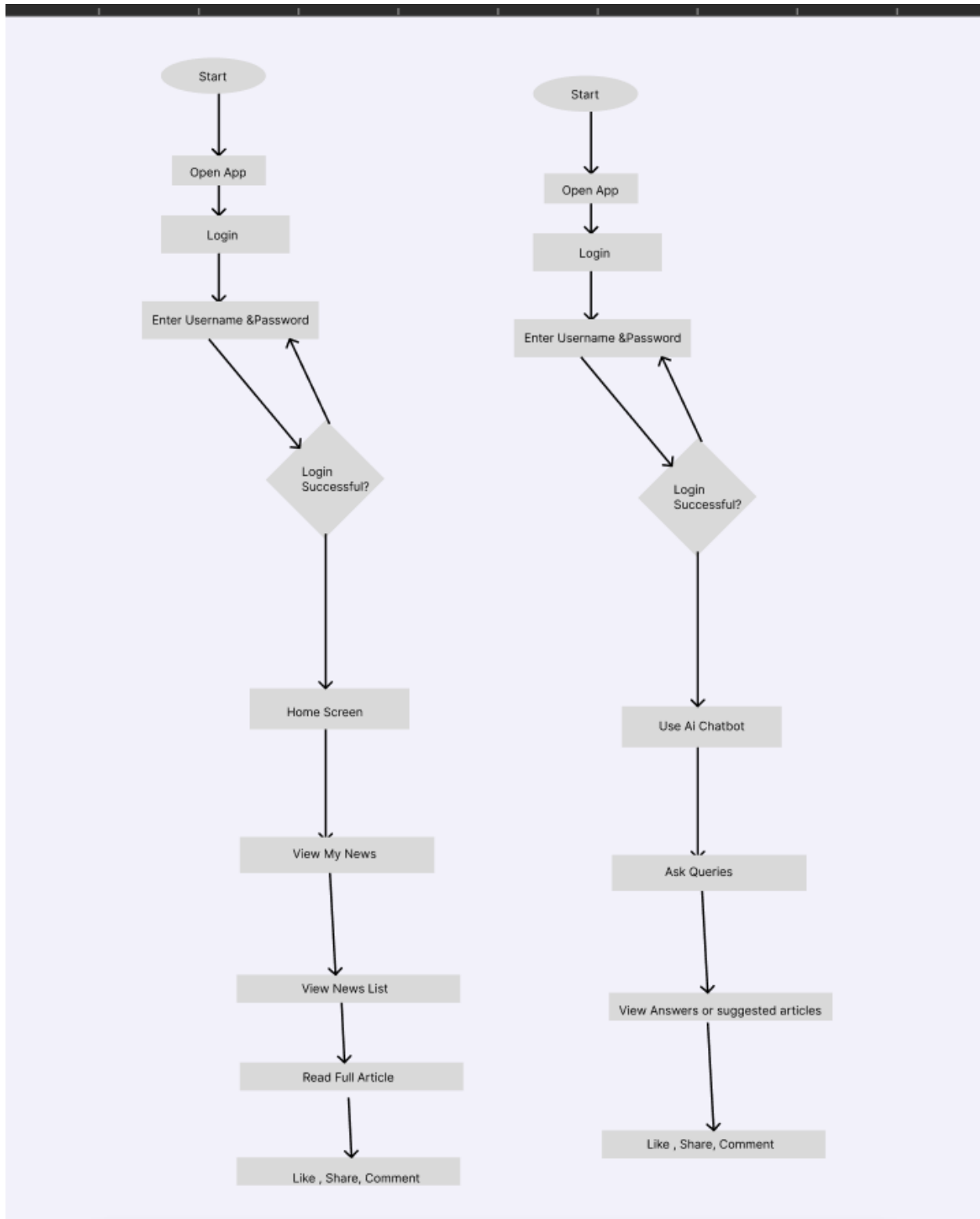


Figure 24 TaskFlow of E patrika

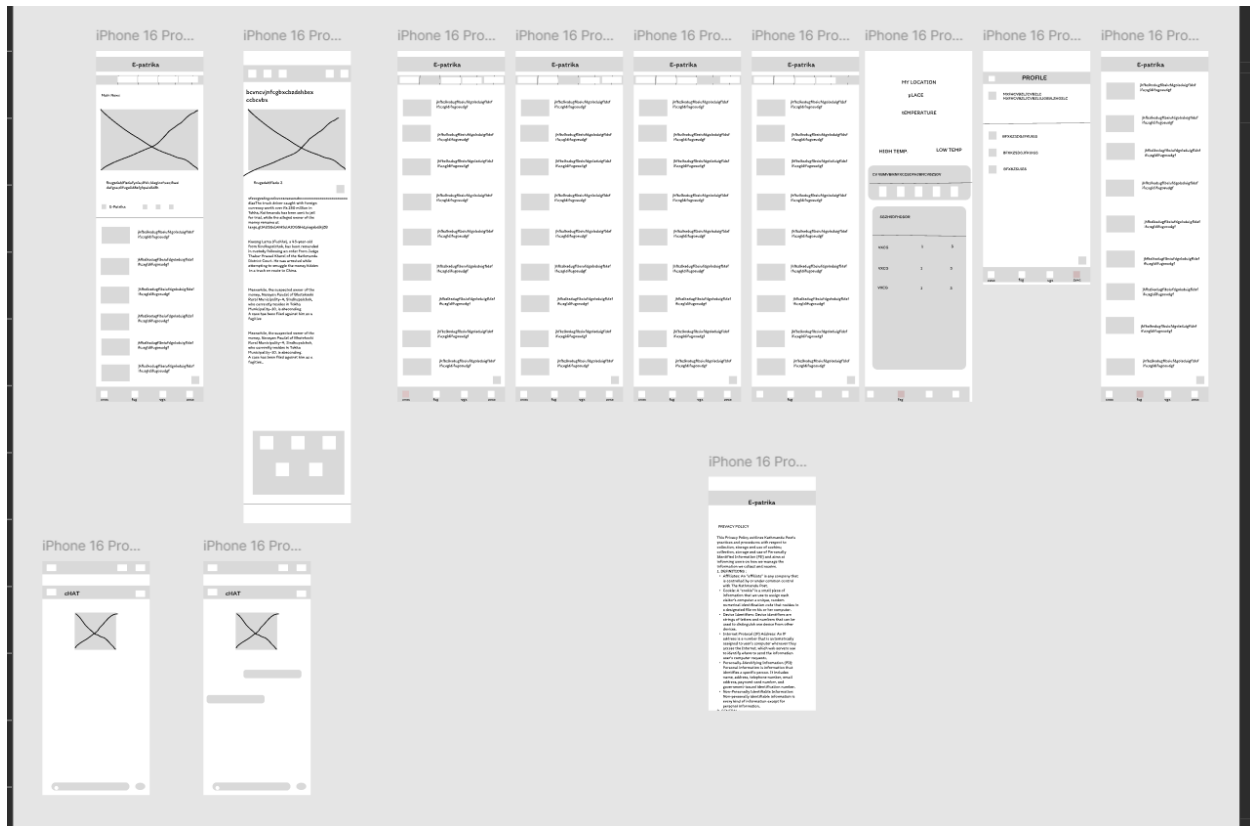


Figure 25 Wireflow of E Patrika

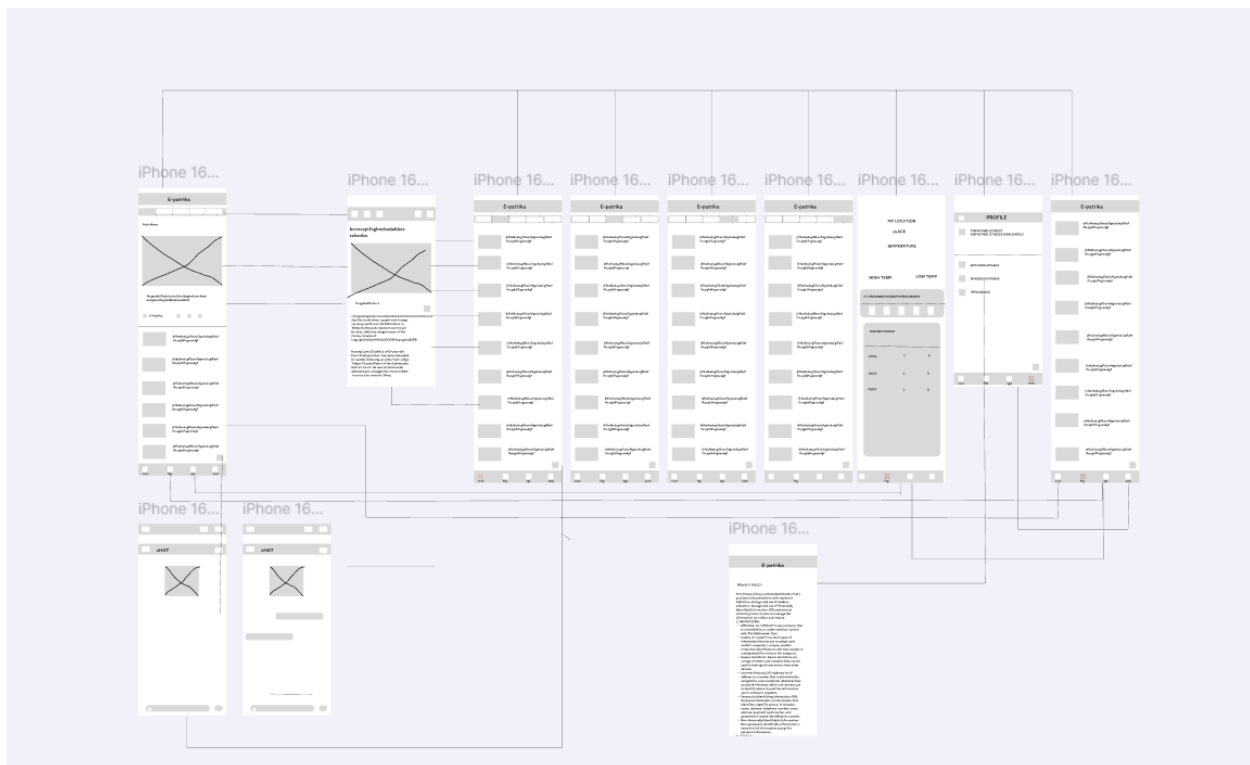


Figure 26 Wireflow of EPatrika

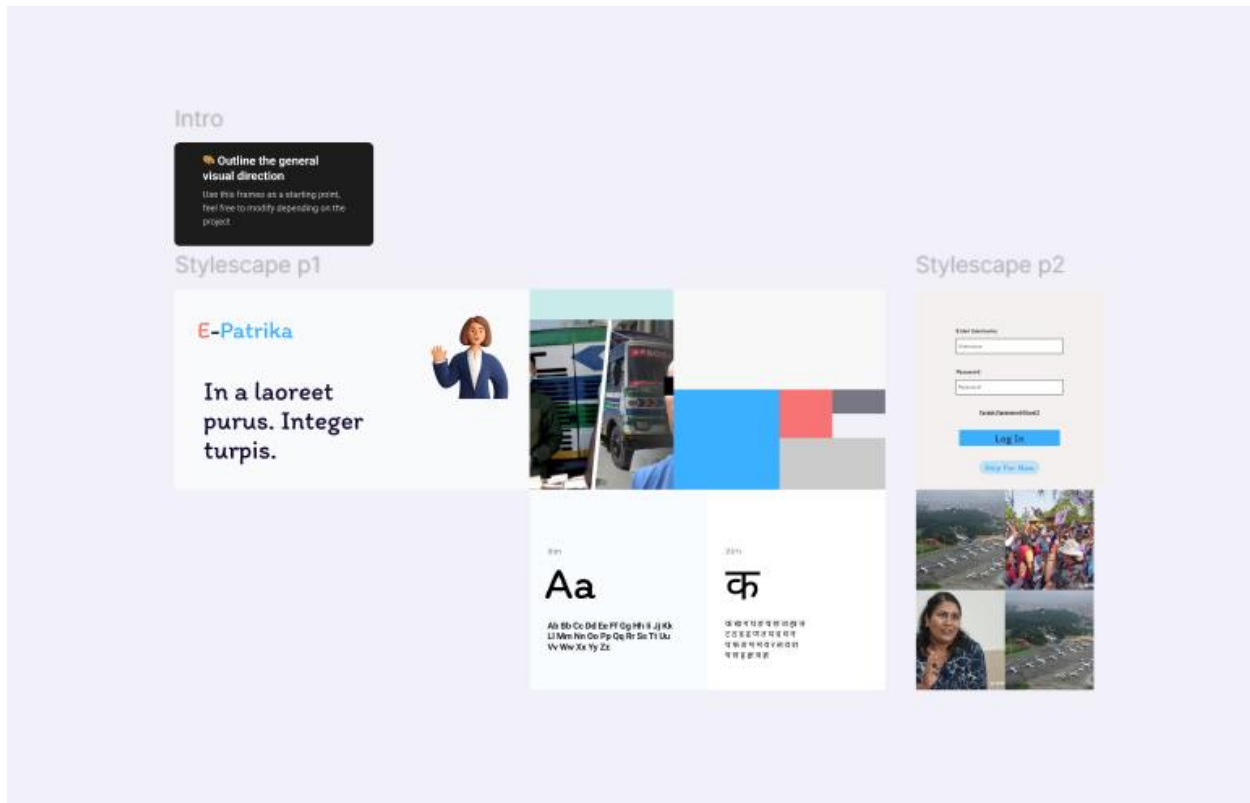


Figure 27Mood of E Patrika

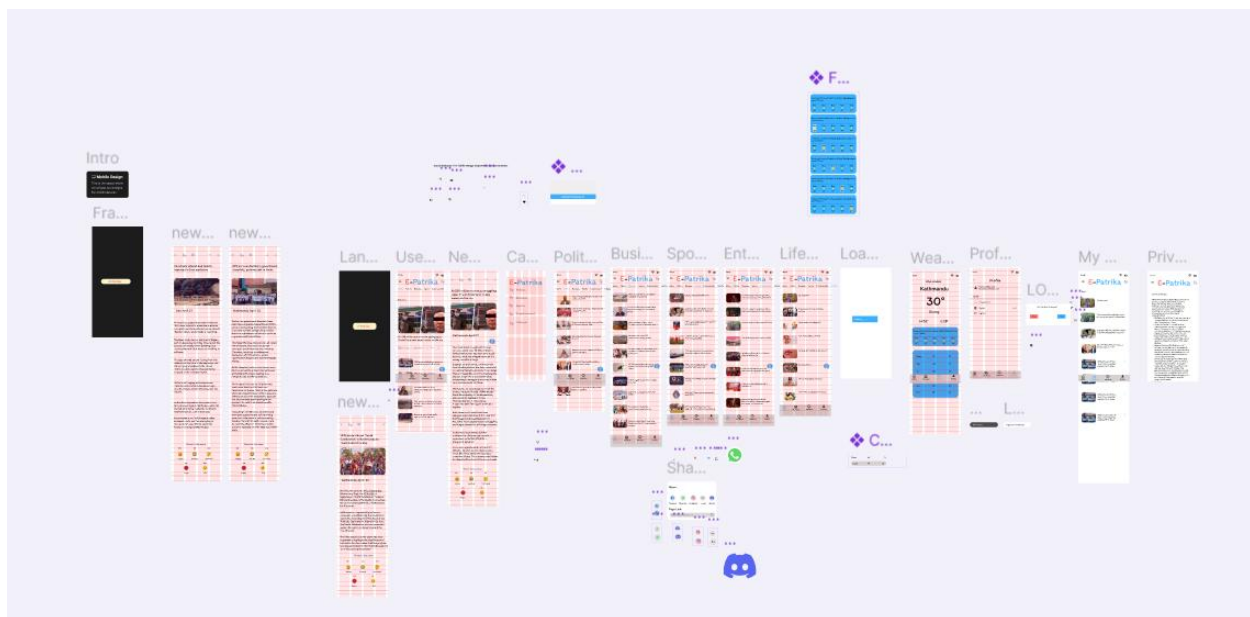


Figure 28Design Process -1

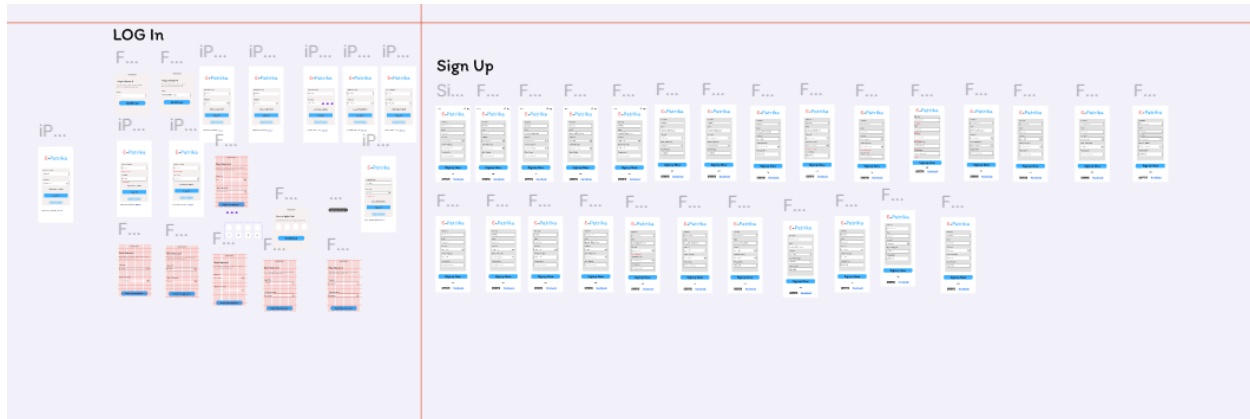


Figure 29 Design Process-2

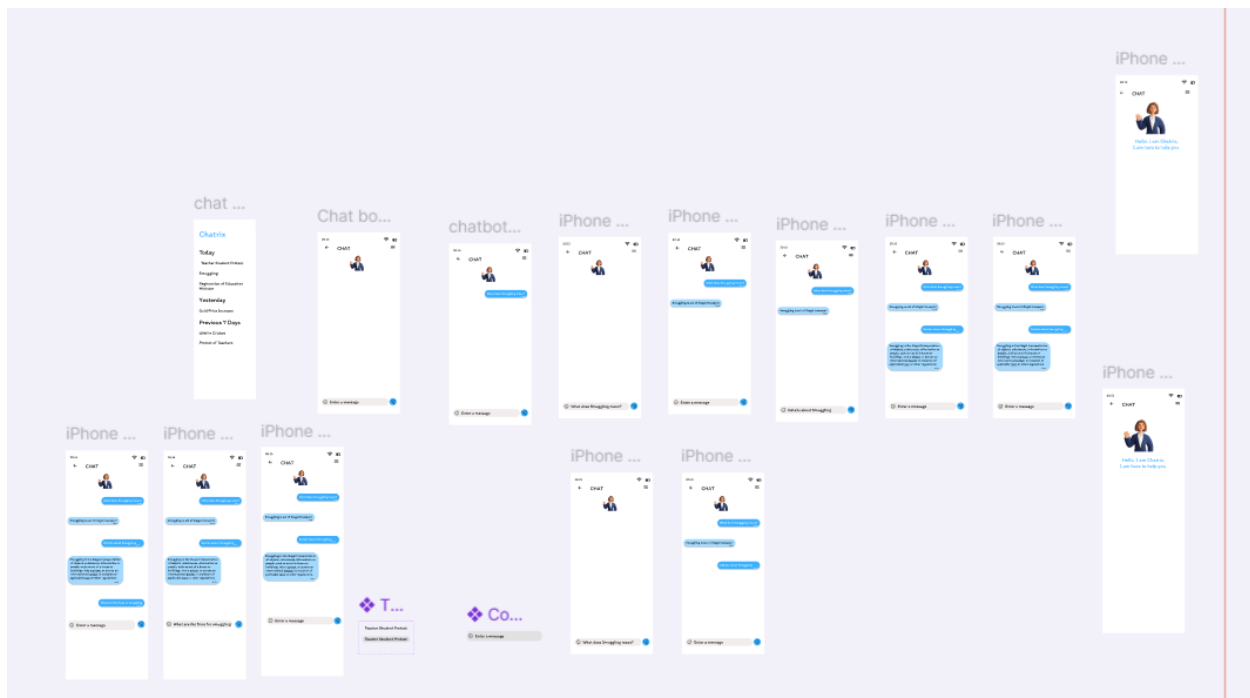


Figure 30 Design Process -3

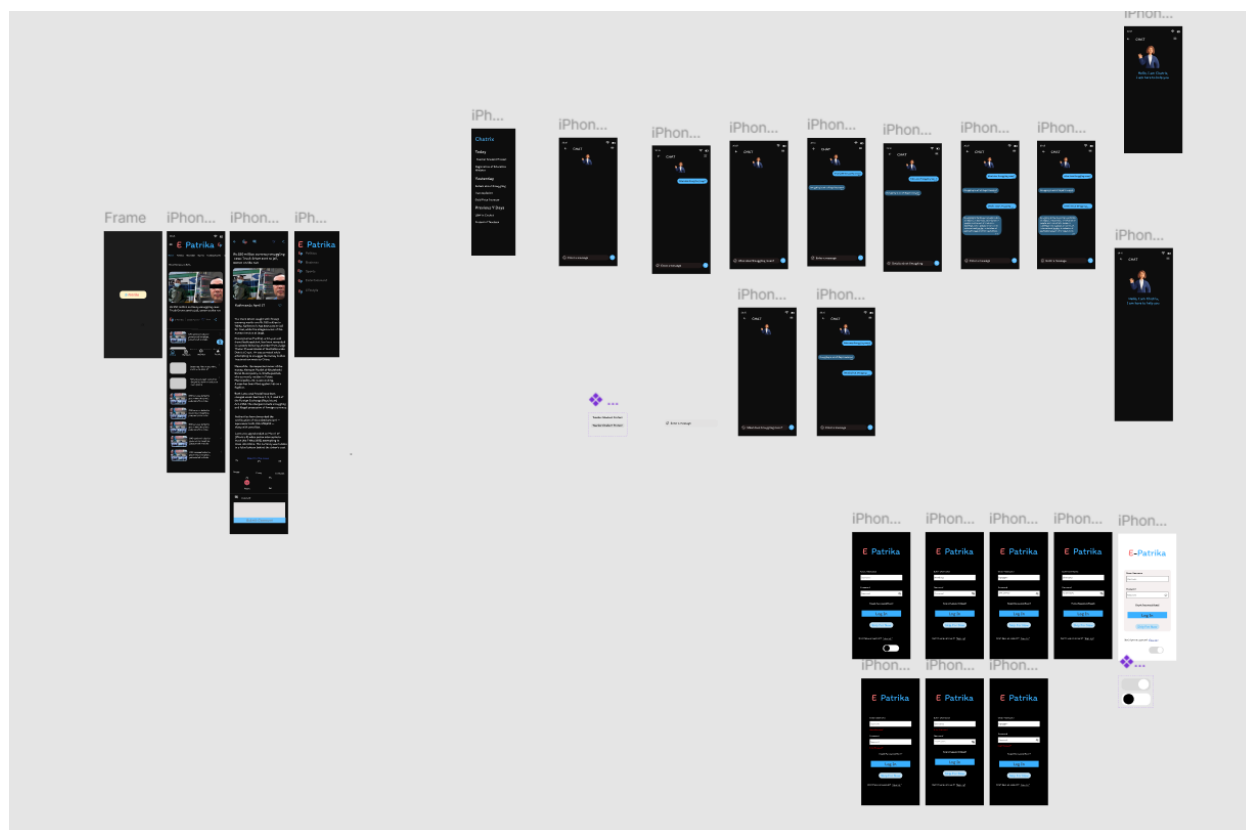


Figure 31 Accessibility Feature 1

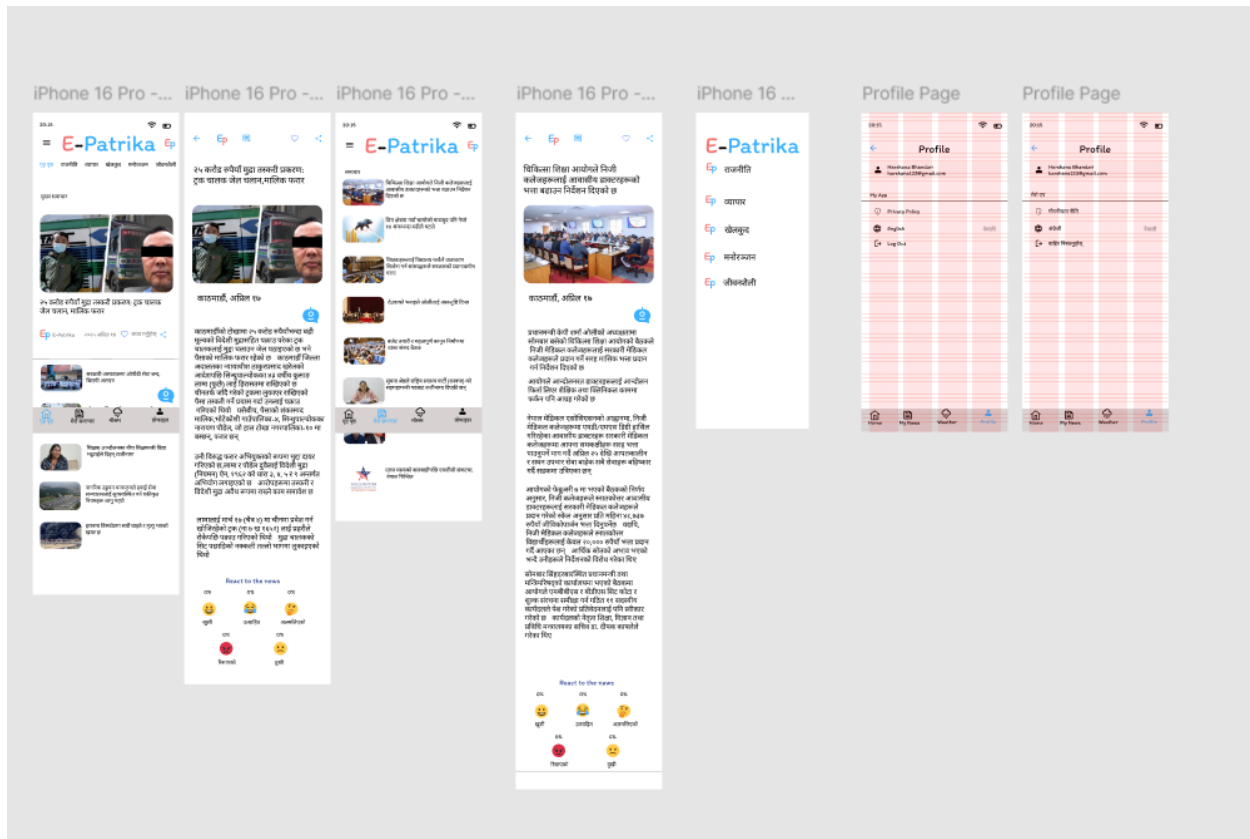


Figure 32 Accessibility Feature 2

5.2 Video Link

<https://drive.google.com/drive/folders/1bhKSFz8VZxSPcVOznqxMdbnBJpVkFmrA>

5.3 Figma Link

[Brainstroming](#)

[User Persona](#)

[Userflow](#)

[Taskflow](#)

[Wireframes](#)

[Wireflow](#)

[Moodboard](#)

[Design](#)

